[SAGE] Training Smart Any-Horizon Agents for Long Video Reasoning with Reinforcement Learning

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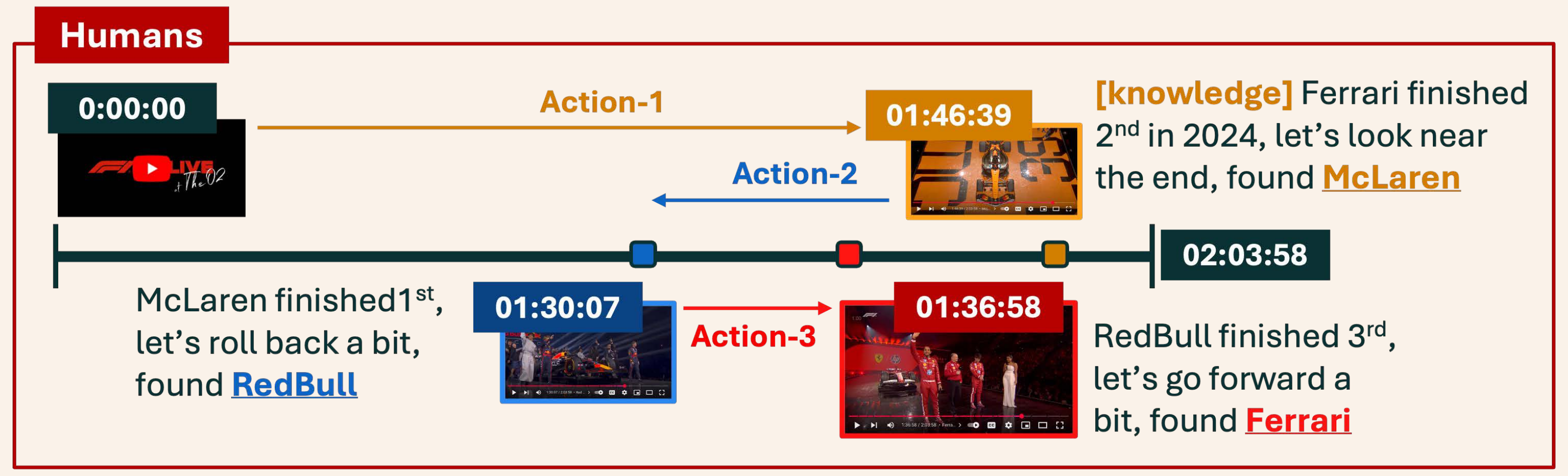
Story of a Formula-1 fan……

*watches Livery reveal event for a Formula 1 2025 season

*wants to know about Ferrari's livery only

*does not want to watch the complete 2-hour video



[knowledge] Ferrari finished 2nd in 2024, let's look near the end, found McLaren

McLaren finished 1st, let's roll back a bit, found RedBull

RedBull finished 3rd, let's go forward a bit, found Ferrari

Humans are natural any-horizon reasoners

Existing Video Reasoning Systems

method	paradigm	mcq	open-ended	runtime (*sec/sample*)
VideoAgent [17]	AGENT	52.6	32.9	1445.9
LVAgent [3]	AGENT	70.5	32.1	92.9
VideoMind [30]	AGENT	69.7	33.2	24.7
VideoExplorer [57]	AGENT	69.6	35.1	137.7
VideoRFT-7B [46]	DIRECT	71.6	41.4	7.2
Video-R1-7B [18]	DIRECT	73.6	43.9	7.3
Qwen3-VL-8B-instruct [41]	DIRECT	77.7	54.0	3.6
Gemini-2.5-Flash [40]	DIRECT	77.2	60.4	—
SAGE (ours)	AGENT	**82.6**	**55.6**	8.6

Open-source video agent systems

underperform on open-ended questions

are slow for real-world use cases *(entertainment)*

*performance on SAGE-Bench

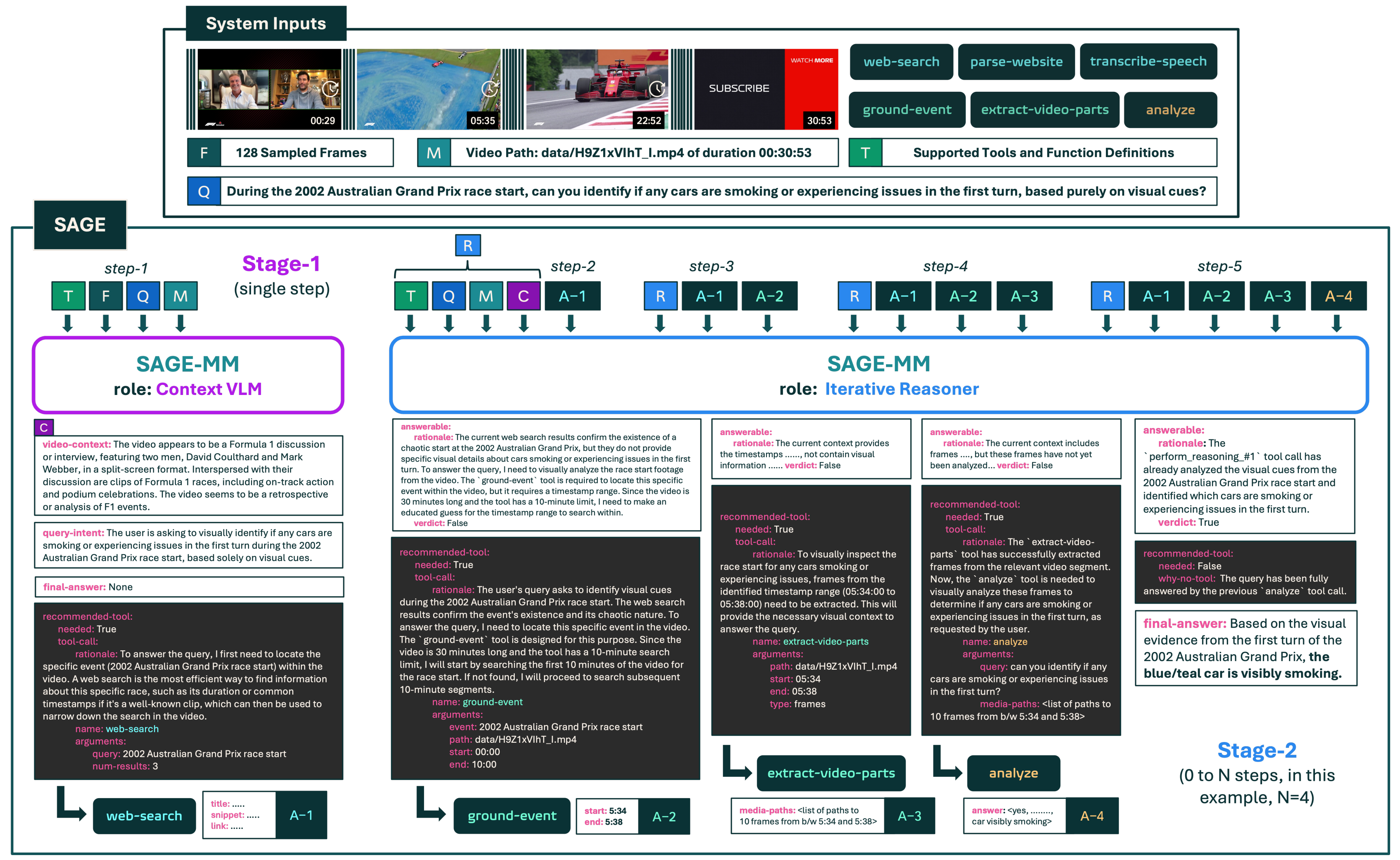
DIRECT	output the final answer with a single sequence prediction process *i.e.*, **single-turn reasoning**.
AGENT	leverage tools to gain more context about the video through **multi-turn reasoning**, ending with an answer turn.

What are the technical challenges toward effectively training video reasoning models under the AGENT paradigm?

[C1] System Design

Existing systems **rely solely on temporal grounding**, ineffective for long videos.

Equip the system with tools like **web-search** and **transcribe-speech** to leverage non-visual signals.



F 128 Sampled Frames

M Video Path: data/H9Z1xVIhT_I.mp4 of duration 00:30:53

T Supported Tools and Function Definitions

Q During the 2002 Australian Grand Prix race start, can you identify if any cars are smoking or experiencing issues in the first turn, based purely on visual cues?

C

video-context: The video appears to be a Formula 1 discussion or interview, featuring two men, David Coulthard and Mark Webber, in a split-screen format. Interspersed with their discussion are clips of Formula 1 races, including on-track action and podium celebrations. The video seems to be a retrospective or analysis of F1 events.

query-intent: The user is asking to visually identify if any cars are smoking or experiencing issues in the first turn during the 2002 Australian Grand Prix race start, based solely on visual cues.

final-answer: None

```
recommended-tool:
    needed: True
    tool-call:
        rationale: To answer the query, I first need to locate the specific event (2002 Australian Grand Prix race start) within the video. A web search is the most efficient way to find information about this specific race, such as its duration or common timestamps if it's a well-known clip, which can then be used to narrow down the search in the video.
        name: web-search
        arguments:
            query: 2002 Australian Grand Prix race start
            num-results: 3
```

web-search → title: ….. snippet: ….. link: ….. A-1

answerable:
rationale: The current web search results confirm the existence of a chaotic start at the 2002 Australian Grand Prix, but they do not provide specific visual details about cars smoking or experiencing issues in the first turn. To answer the query, I need to visually analyze the race start footage from the video. The `ground-event` tool is required to locate this specific event within the video, but it requires a timestamp range. Since the video is 30 minutes long and the tool has a 10-minute limit, I need to make an educated guess for the timestamp range to search within.
verdict: False

```
recommended-tool:
    needed: True
    tool-call:
        rationale: The user's query asks to identify visual cues during the 2002 Australian Grand Prix race start. The web search results confirm the event's existence and its chaotic nature. To answer the query, I need to locate this specific event in the video. The `ground-event` tool is designed for this purpose. Since the video is 30 minutes long and the tool has a 10-minute search limit, I will start by searching the first 10 minutes of the video for the race start. If not found, I will proceed to search subsequent 10-minute segments.
        name: ground-event
        arguments:
            event: 2002 Australian Grand Prix race start
            path: data/H9Z1xVIhT_I.mp4
            start: 00:00
            end: 10:00
```

ground-event → start: 5:34 end: 5:38 A-2

answerable:
rationale: The current context provides the timestamps ……, not contain visual information …… **verdict:** False

```
recommended-tool:
    needed: True
    tool-call:
        rationale: To visually inspect the race start for any cars smoking or experiencing issues, frames from the identified timestamp range (05:34:00 to 05:38:00) need to be extracted. This will provide the necessary visual context to answer the query.
        name: extract-video-parts
        arguments:
            path: data/H9Z1xVIhT_I.mp4
            start: 05:34
            end: 05:38
            type: frames
```

extract-video-parts → media-paths: <list of paths to 10 frames from b/w 5:34 and 5:38> A-3

answerable:
rationale: The current context includes frames …., but these frames have not yet been analyzed… **verdict:** False

```
recommended-tool:
    needed: True
    tool-call:
        rationale: The `extract-video-parts` tool has successfully extracted frames from the relevant video segment. Now, the `analyze` tool is needed to visually analyze these frames to determine if any cars are smoking or experiencing issues in the first turn, as requested by the user.
        name: analyze
        arguments:
            query: can you identify if any cars are smoking or experiencing issues in the first turn?
            media-paths: <list of paths to 10 frames from b/w 5:34 and 5:38>
```

analyze → answer: <yes, …….., car visibly smoking> A-4

answerable:
rationale: The `perform_reasoning_#1` tool call has already analyzed the visual cues from the 2002 Australian Grand Prix race start and identified which cars are smoking or experiencing issues in the first turn.
verdict: True

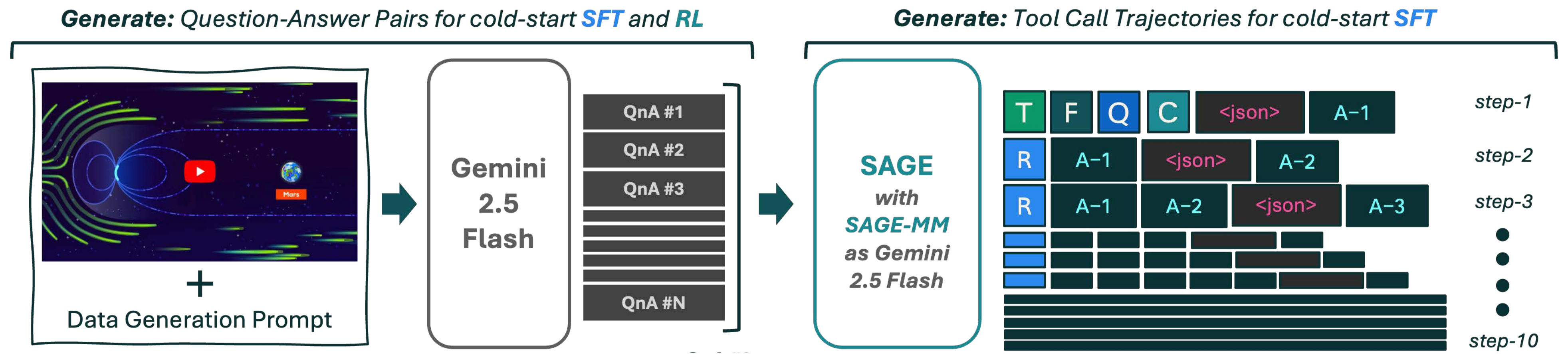
```
recommended-tool:
    needed: False
    why-no-tool: The query has been fully answered by the previous `analyze` tool call.
```

final-answer: Based on the visual evidence from the first turn of the 2002 Australian Grand Prix, **the blue/teal car is visibly smoking.**

[C2] Data

Training requires QnA pairs and **human annotation is expensive** for long videos.

Cost-effective synthetic data pipeline using Gemini-2.5-Flash.


Generate: Question-Answer Pairs for cold-start SFT and RL
Mars
+
Data Generation Prompt
Gemini 2.5 Flash
QnA #1
QnA #2
QnA #3
QnA #N
Generate: Tool Call Trajectories for cold-start SFT
SAGE with SAGE-MM as Gemini 2.5 Flash
T
F
Q
C
<json>
A-1
R
A-2
A-3
step-1
step-2
step-3
step-10

[C3] Adaptive Reasoning

Hard to train models for any-horizon behavior due to **variable video durations.**

A multi-reward RL recipe using a **strong reasoning LLM as a judge** with process and outcome rewards.

[process]

$$s_{\text{format}} = \begin{cases} +0.05 & \textit{JSON contains only required fields} \\ -0.10 & \textit{otherwise} \end{cases}$$

$$s_{\text{args-valid}} = \begin{cases} -0.10 & \textit{arguments are invalid} \\ 0 & \textit{otherwise} \end{cases}$$

$$s_{\text{args-repeat}} = -0.05 \times \sqrt{(\text{num-repetitions})}$$

$$s_{\text{reasonable-tool}} = \begin{cases} +0.10 & \textit{tool call reasonable } \textbf{(GPT-4o judge)} \\ -0.10 & \textit{otherwise} \end{cases}$$

[outcome]

$$a_n = \begin{cases} -2.0 & \textit{JSON action string is invalid} \\ -0.5 & \textit{wrong answer, } N > 1 \textbf{ [any-horizon penalty]} \\ +1.25 & \textit{correct + } \textbf{visual tools} \textit{ used in } \tau_i \\ +1.0 & \textit{correct answer (direct or non-visual tools)} \end{cases}$$

$$R_i = (s_1 + s_2 + \dots + s_n) + a_n$$

$$r(A_1) = r(A_2) = \dots = r(A_n) = R_i$$

synchronous RL

Results

Performance improvements and number of turns gradually grow as video duration increases **[any-horizon behavior]**

Method	Model	Eval Mode	0-60	60-180	180-300	300-600	600-1200	1200-2400	2400+	overall
			(261)	(390)	(116)	(186)	(484)	(147)	(180)	(1744)
Qwen3-VL (baseline)	Qwen3-VL-8B-Instruct	DIRECT	73.9	72.3	**81.9**	71.5	55.0	59.2	47.5	64.9
SAGE (ours)	Qwen3-VL-8B-Instruct [+SFT]	AGENT	74.3	68.1	75.0	72.0	56.8	55.8	48.1	63.9
SAGE (ours)	**SAGE-MM**: Qwen3-VL-8B-Instruct [+SFT] [+RL]	AGENT	**78.5** (+4.6)	70.3 (-2.0)	77.4 (-4.5)	**72.6** (+1.1)	**63.2** (+8.2)	**61.9** (+2.7)	**53.8** (+6.3)	**68.1** (+3.2)
SAGE-Flash (ours)	**SAGE-MM**: Qwen3-VL-8B-Instruct [+SFT] [+RL]	AGENT	**77.8** (+3.9)	**73.6** (+1.3)	80.2 (-1.7)	**76.3** (+4.8)	**69.6** (+14.6)	**68.0** (+8.8)	**56.2** (+8.7)	**71.8** (+6.9)

SAGE-MM	**0-60**	**60-180**	**180-300**	**300-600**	**600-1200**	**1200-2400**	**2400+**
Qwen3-VL-8B-Instruct [+SFT]	2.00	2.23	2.05	2.63	3.02	3.50	3.54
Qwen3-VL-8B-Instruct [+SFT] [+RL]	1.74	1.81	1.83	2.18	2.49	2.89	2.77

	SAGE-MM	Video-MMMU	Video-MME
Qwen2.5-VL-7B-Instruct [2]	N/A	57.5	63.6
Qwen3-VL-8B-Instruct [41]	N/A	65.3	**66.8**
Video-R1-7B [18]	N/A	61.5	61.2
SAGE-Flash (ours)	Qwen3-VL-8B-Instruct [+SFT]	66.9	59.4
SAGE-Flash (ours)	Qwen3-VL-8B-Instruct [+SFT] [+RL]	**68.1**	63.5
- w/o ground-event		65.8	65.6
- w/o ground-event & extract-video-parts		61.8	66.2

Generalization to knowledge acquisition from videos.

Temporal grounding can still be a bottleneck.

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